

Decent Coaching Classes

VIII CBSE

Mind Your Math - V

ANSWER KEY

Marks: 30

Time: 1 hr.

Section A – Multiple Choice Questions (1 mark each)

[6]

1. Answer: (B) 6. [The coefficient of a^2b is the number multiplying it in the first term, i.e. 6.]
2. Answer: (C) $x^2 + 5\sqrt{x} - 7$. [$\sqrt{x} = x^{1/2}$; the power of the variable is not a whole number, so it is not a polynomial.]
3. Answer: (D) trinomial. [It has three unlike terms : $7ab$, $-5a$ and $3b^2$.]
4. Answer: (A) $4x^3 - 12xy$. [$4x \times x^2 = 4x^3$; $4x \times (-3y) = -12xy$]
5. Answer: (B) $7ab^2$ and $-3ab^2$. [Same variables with the same powers; only the coefficients differ.]
6. Answer: (C) $x^2 + x - 6$. [$(x+3)(x-2) = x^2 - 2x + 3x - 6 = x^2 + x - 6$]

Section B – Very Short Answer Questions (2 marks each)

[10]

7. Coefficient and constant term
Coefficient of $xy = -5$ (1 mark)
Constant term = 9 (1 mark)
8. Classification
(i) $6xy$ has one term, so it is a monomial. (1 mark)
(ii) $a^2 + a - 5$ has three terms, so it is a trinomial. (1 mark)
9. Like terms
Like terms : $3xy, -7xy, xy$ (1 mark)
Like terms : $5x^2, 4x^2$ (1 mark) [2x has no like term in the list.]
10. Find the product : $\frac{3}{4}a(8a^2 - 4ab)$
 $= (\frac{3}{4}a) \times 8a^2 - (\frac{3}{4}a) \times 4ab$ (1 mark)
 $= 6a^3 - 3a^2b$ (1 mark)
11. Subtract $5x - 3xy + 2$ from $8x + xy - 6$
 $(8x + xy - 6) - (5x - 3xy + 2) = 8x + xy - 6 - 5x + 3xy - 2$ (1 mark)
 $= (8x - 5x) + (xy + 3xy) + (-6 - 2)$
 $= 3x + 4xy - 8$ (1 mark)

Section C – Short Answer Questions (3 marks each)

[9]

12. Simplify : $3x(x - 2) + x(5 - 4x) + 2x^2$
 $= (3x^2 - 6x) + (5x - 4x^2) + 2x^2$ (1 mark for opening the brackets)
 $= (3x^2 - 4x^2 + 2x^2) + (-6x + 5x)$ (1 mark for grouping like terms)
 $= x^2 - x$ (1 mark)
13. Multiply $(3a - 5)$ and $(2a + 7)$. Verify for $a = 1$.
 $(3a - 5)(2a + 7) = 3a(2a + 7) - 5(2a + 7)$
 $= 6a^2 + 21a - 10a - 35$
 $= 6a^2 + 11a - 35$ (2 marks)
Verification for $a = 1$:
LHS = $(3 \times 1 - 5)(2 \times 1 + 7) = (-2)(9) = -18$
RHS = $6(1)^2 + 11(1) - 35 = 6 + 11 - 35 = -18$
LHS = RHS, hence verified. (1 mark)
14. (a) $(2a + b)(4a^2 - 2ab + b^2)$

$$= 2a(4a^2 - 2ab + b^2) + b(4a^2 - 2ab + b^2) \quad (1 \text{ mark})$$

$$= 8a^3 - 4a^2b + 2ab^2 + 4a^2b - 2ab^2 + b^3 \quad (1 \text{ mark})$$

$$= 8a^3 + b^3 \quad (1 \text{ mark})$$

OR

$$(b) (a^2 - 3a + 5)(2a + 1)$$

$$= a^2(2a + 1) - 3a(2a + 1) + 5(2a + 1) \quad (1 \text{ mark})$$

$$= 2a^3 + a^2 - 6a^2 - 3a + 10a + 5 \quad (1 \text{ mark})$$

$$= 2a^3 - 5a^2 + 7a + 5 \quad (1 \text{ mark})$$

Section D – Long Answer Question (5 marks)

[5]

15. Area of the rectangular garden

$$\text{Area} = \text{length} \times \text{breadth} = (3x + 2)(2x - 1) \quad (1 \text{ mark})$$

$$= 3x(2x - 1) + 2(2x - 1)$$

$$= 6x^2 - 3x + 4x - 2 \quad (1 \text{ mark})$$

$$\text{Area} = (6x^2 + x - 2) \text{ square metres} \quad (1 \text{ mark})$$

Verification for $x = 3$:

$$\text{Length} = 3(3) + 2 = 11 \text{ m, Breadth} = 2(3) - 1 = 5 \text{ m, so Area} = 11 \times 5 = 55 \text{ m}^2$$

$$\text{Using the result : } 6(3)^2 + 3 - 2 = 54 + 3 - 2 = 55 \text{ m}^2 \quad (1 \text{ mark})$$

Both values are equal, hence verified. (1 mark)

OR

$$\text{Simplify : } (2x - 3)(3x + 2) - (4x + 1)(x - 2)$$

$$(2x - 3)(3x + 2) = 6x^2 + 4x - 9x - 6 = 6x^2 - 5x - 6 \quad (1 \text{ mark})$$

$$(4x + 1)(x - 2) = 4x^2 - 8x + x - 2 = 4x^2 - 7x - 2 \quad (1 \text{ mark})$$

$$\text{Difference} = (6x^2 - 5x - 6) - (4x^2 - 7x - 2) = 6x^2 - 5x - 6 - 4x^2 + 7x + 2$$

$$= 2x^2 + 2x - 4 \quad (1 \text{ mark})$$

Verification for $x = 2$:

$$\text{LHS} = (4 - 3)(6 + 2) - (8 + 1)(2 - 2) = (1)(8) - (9)(0) = 8$$

$$\text{RHS} = 2(2)^2 + 2(2) - 4 = 8 + 4 - 4 = 8 \quad (1 \text{ mark})$$

LHS = RHS, hence verified. (1 mark)